Interpreting Models: Decision Trees & Gini

GeoPython — Quick Reference Handout 07

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1. Threshold-Based Science

In geoscience, we are not always looking for a complex equation. Sometimes the most valuable output is a clear threshold — a boundary value the tree identifies automatically — that signals a meaningful shift in system behaviour.

A decision tree finds these thresholds automatically by repeatedly asking binary questions about the data. Each split divides the dataset into two groups, attempting to make each group as pure (same-class) as possible.

> **!** *A threshold discovered by a decision tree is a scientific hypothesis. Ask: Is this value physically meaningful? Does it correspond to a known transition (e.g., a freeze/thaw threshold, a phenological boundary)?*

2. Gini Impurity: Measuring Purity

Gini impurity measures how "mixed" a node is. A score of 0.0 means the node is perfectly pure — every sample belongs to the same class. The tree algorithm chooses splits that minimise Gini.

Gini Score	Meaning	Example
0.0	Perfectly pure — one class only	All samples: "melt season"
0.5	Maximum impurity (two equal classes)	Half melt, half freeze — no predictive power
0.0–0.2	Useful leaf — high confidence prediction	Aim for this at terminal nodes
> 0.4	Weak split — consider pruning	Node adds little discriminating power

3. Dataset Agnosticism

One of the most important properties of decision trees is that they make no assumptions about the distribution of your data. They are dataset agnostic — they work on temperatures, ice extents, soil carbon, rainfall anomalies, or categorical labels without modification.

Property	Decision Tree Behaviour
Scaling required?	No — splits are ordinal, not distance-based
Handles mixed data types?	Yes — numeric and categorical in same tree
Interpretable?	Yes — you can draw and read every decision
Prone to overfitting?	Yes — limit max_depth or use min_samples_leaf
Best ensemble method?	Random Forest (many trees, averaged predictions)

> *Always visualise your tree (sklearn.tree.plot_tree). A tree that is too deep memorises noise. A tree with depth 3–5 is usually the most interpretable and generalisable for geoscience datasets.*